

Group number: _____ Date: _____

Date:

Air Quality Science: Lab Worksheet

UTSC Event for Visiting Students

“Best” AQ measurements and location:

“Worst” AQ measurements and location:

AQ measurements at AA building + validation measurement:

Data log

Consider noting both the typical (mean) value and its variability using $\pm$,

e.g., 425 ± 15 ppm would mean it varied between 410 and 440 ppm.

[illegible]

Data continued:

Time	Location	PM2.5 [$\mu\text{g}/\text{m}^3$]	CO ₂ [ppm]	HCHO [mg/m ³]	Notes

Specific Questions and Guidance

Before leaving, consider:

- Where do you expect to see high vs. low particulate matter?
- Where do you expect to see high vs. low CO₂?

During campus measurements:

- Where is the 'best' and where is the 'worst' air quality on campus? Start with your expectations (hypothesis) and compare to your measurements.
 - Note: I am not defining 'best' and 'worst': consider how you would define these terms and why.
- How does indoor CO₂ compare to outdoor CO₂? Are all indoor and outdoor areas the same? What accounts for the differences?
- Are there any patterns in PM_{2.5} levels? What factors seemed to influence your observations?
- How much did the values fluctuate while staying in one location of interest for a few minutes? What might explain this short-term variability?
- Decide in advance on one to three highlights you might want to discuss with the class once you return, e.g., an unexpected observation, a high-pollution (or low-pollution) location, or an indoor/outdoor difference.

After returning to lecture room and reflecting on your measurements:

- Did your observations about the 'best' and 'worst' locations for CO₂ and PM2.5 align with your expectations? Explain.
- Were there any surprising or contradictory results?
- If we examine the measurements of PM2.5 taken at the AA roof by a higher quality instrument, do you expect they will match your measurements?
 - What are possible reasons for discrepancies?
- If you were designing an air quality monitoring system for a city, how would you use these sensors, or sensors like it? How would you decide where to put them?